

Portal Hypertension, Noncirrhotic, 2 (PHNC2 / NCPH2) — Comprehensive Disease Report

Disease: Portal Hypertension, Noncirrhotic, 2 **MONDO:** MONDO:0030397 · **OMIM:** #619463 · **Causal gene:** *GIMAP5* (HGNC:18005) **Evidence base:** Aggregated disease-level resources (OMIM, MONDO, HPO, gnomAD, UniProt) plus individual-patient reports (~21 patients published to date). Fewer than ~25 molecularly confirmed patients worldwide; all conclusions rest on small case series, a landmark human+mouse mechanistic study, and rodent models.

Summary (Answer to the Research Question)

Portal Hypertension, Noncirrhotic, 2 (PHNC2) is a **rare autosomal-recessive disorder caused by biallelic loss-of-function variants in *GIMAP5***, a small organellar GTPase of the immunity-associated protein (IAN) family expressed selectively in **lymphocytes and endothelial cells**. Loss of *GIMAP5* produces a **two-arm disease**: (1) a **hepatic vascular arm** in which *GIMAP5*-deficient liver sinusoidal endothelial cells (LSECs) fail to maintain their identity (via reduced GATA4), undergo **capillarization**, and cause **noncirrhotic (porto-sinusoidal) portal hypertension**; and (2) an **immune arm** with lymphopenia, autoimmune cytopenias, and recurrent infections. A unifying biochemical lesion is **pathological accumulation of long-chain ceramides** (*GIMAP5* normally restrains CK2-driven ceramide synthase activity), driving cellular senescence in both endothelium and lymphocytes. Presentation is typically in **childhood** with splenomegaly, thrombocytopenia, esophageal varices, and elevated transaminases, without cirrhosis.

1. Disease Information

Overview. PHNC2 is a Mendelian cause of **idiopathic/noncirrhotic portal hypertension**, now classified within the spectrum of **porto-sinusoidal vascular disorder (PSVD)**. It is distinctive because portal hypertension arises from a primary **endothelial** defect rather than from hepatocellular injury or cirrhosis, and it co-occurs with an **inborn error of immunity**.

Key identifiers - **MONDO:** MONDO:0030397 ("portal hypertension, noncirrhotic, 2"; synonym **NCPH2**) - **OMIM:** #619463 (phenotype) - **UMLS:** C5561948 · **MedGen:** 1794158 - **Gene:** *GIMAP5* — HGNC:18005; NCBI Gene 55340; Ensembl ENSG00000196329; UniProt **Q96F15**; gene **MIM 608086**; cytoband **7q36.1** - **ICD-10:** K76.6 (portal hypertension; no PHNC2-specific code) · **ICD-11:** DB98.5 region (non-cirrhotic portal hypertension/idiopathic; no specific code) - **MeSH:** most closely "Hypertension, Portal" (D006975); no PHNC2-specific MeSH - **Orphanet:** no PHNC2-specific ORPHA; overlaps "Idiopathic non-cirrhotic portal hypertension"/PSVD

Synonyms / alternative names: NCPH2; GIMAP5 deficiency; GIMAP5-related noncirrhotic portal hypertension; GIMAP5-related porto-sinusoidal vascular disorder; (historically discussed with "idiopathic noncirrhotic portal hypertension," INCPH). Note: **PHNC1 (INCPH1, OMIM 617068)** is a distinct entity caused by *DGUOK* (P 26874653).

Data provenance: Disease-level ontologies/aggregators + individual patient case reports/series (human clinical), complemented by mouse and rat model data and in-vitro mechanistic studies.

2. Etiology

Primary cause — genetic (monogenic). Biallelic (homozygous or compound-heterozygous) **loss-of-function variants in *GIMAP5***. *"we demonstrate homozygous damaging mutations in GIMAP5, a small organellar GTPase, in four families with unexplained portal hypertension"* (Drzewiecki et al., P 33956074).

Genetic risk factors - Causal variants: recessive *GIMAP5* alleles (see §4). Disease requires **two damaged alleles**. - **Consanguinity** is a major contributor: three of four index families were consanguineous (first-cousin unions).

([P 33956074](#))

- **Founder/recurrent allele: p.Leu204Pro** recurs across independent reports and is relatively common (gnomAD AF $\approx 2.1 \times 10^{-3}$), so it is the most likely allele to appear in homozygous or compound-heterozygous state.

Environmental risk factors. None established. No toxin, infection, diet, or occupational exposure is required for disease; the phenotype is genetically determined. (Of note, acquired phenocopies of noncirrhotic portal hypertension exist — e.g., didanosine/thioguanine exposure, oxaliplatin — but these are not PHNC2.)

Protective factors. No genetic or environmental protective factors are defined. At the population level *GIMAP5* is **LOF-tolerant** (gnomAD pLI = 0.003; observed/expected LOF = 8/9.9), so **monoallelic carriers are generally healthy** — heterozygosity is effectively "protective" relative to the biallelic state.

Gene–environment interactions. A 2026 report raised the hypothesis that **partial (monoallelic) *GIMAP5* deficiency might subtly predispose to localized vascular anomalies under specific environmental/epigenetic conditions via multi-hit mechanisms** — the dizygotic twin carrying a single p.Leu204Pro allele showed a localized fibro-adipose vascular anomaly but no immune defect.

([P 42358996](#))

This remains speculative.

3. Phenotypes (with HPO terms, type, and frequency)

Onset is typically in **infancy/childhood**; severity is **variable**; the hepatic disease is **chronic and progressive**.

Phenotype	HPO	Type	Frequency / notes
Portal hypertension	HP:0001409	Clinical sign	Defining; universal
Splenomegaly	HP:0001744	Physical sign	All affected subjects (P 33956074)
Thrombocytopenia	HP:0001873	Lab abnormality	Near-universal (hypersplenism)
Elevated transaminases	HP:0002910	Lab abnormality	Near-universal
Esophageal varices	HP:0002040	Clinical sign	Near-universal; 7 subjects endoscopically confirmed
Nodular regenerative hyperplasia of liver	HP:0011954	Pathology	Present on biopsy
Hepatomegaly	HP:0002240	Physical sign	Common
Elevated GGT	HP:0030948	Lab abnormality	Reported
Ascites	HP:0001541	Clinical sign	Complication of portal hypertension
Ecchymosis / petechiae / epistaxis	HP:0031364 / HP:0000967 / HP:0000421	Signs	Bleeding from thrombocytopenia/varices
Hemoptysis	HP:0002105	Sign	Reported
Recurrent infections	HP:0002719	Sign	Immune arm
Autoimmune cytopenias (hemolytic anemia, immune thrombocytopenia)	HP:0001890 / HP:0001873	Lab/clinical	Immune dysregulation
Lymphopenia	HP:0001888	Lab abnormality	T (and NK) lymphopenia; T-cell exhaustion
Hepatocellular carcinoma	HP:0001402	Neoplasm	Listed as possible long-term risk

Phenotype	HPO	Type	Frequency / notes
Fatigue	HP:0012378	Symptom	Nonspecific

Age of onset: neonatal–childhood (pediatric); portal hypertension complications and immune features generally manifest in the first two decades. **Progression:** progressive liver/vascular disease (one patient developed worsening coagulopathy and direct hyperbilirubinemia after shunt surgery).

([P 33956074](#))

Quality-of-life impact: substantial — recurrent variceal bleeding risk, transfusion-dependent cytopenias, infection susceptibility, growth/activity limitation, and the burden of surveillance endoscopy and possible transplantation. No formal EQ-5D/SF-36/PROMIS data exist for this ultra-rare disease.

4. Genetic / Molecular Information

Causal gene: *GIMAP5* (GTPase, IMAP family member 5) — HGNC:18005; NCBI Gene 55340; Ensembl ENSG00000196329; UniProt **Q96F15** (307 aa); gene **MIM 608086**; **7q36.1** (GRCh38 chr7:150,722,253–150,750,033, + strand). Part of the IAN GTPase cluster on 7q36.1; read-through transcript with upstream *GIMAP1* exists. Aliases: IAN5, IAN4L1, IROD, **NCPH2**.

Protein architecture: GIMAP5 is a small **AIG1-type guanine-nucleotide-binding (G) domain** GTPase (InterPro **IPR006703**; Pfam **PF04548** "AIG1 family"; PROSITE PS51720), belonging to the P-loop NTPase superfamily (IPR027417) and the **GTPase GIMA/IAN/Toc** family (IPR045058), with a C-terminal transmembrane anchor targeting it to lysosomal/endosomal membranes. The reported missense substitutions (I47T, P109L, L204P, L223P) map to/near this G-domain and are predicted to **destabilize the fold and abolish GTPase function**, consistent with the observed absent protein for null combinations.

Pathogenic variants (all **germline**, all **loss-of-function / recessive**):

Variant (protein)	cDNA/ type	Zygosity	Population frequency	Source
p.Ile47Thr (I47T)	missense	homozygous (kindred 1)	AF = 0 in gnomAD	P 33956074
p.Pro109Leu (P109L)	missense	homozygous (kindred 3)	ultra-rare (1/251,400 alleles)	P 33956074
p.Leu204Pro (L204P)	missense	homozygous (kindred 4); also compound-het	AF $\approx 2.1 \times 10^{-3}$ (recurrent)	P 33956074
				P 29382851
				P 42358996
p.Leu223Pro (L223P)	missense	homozygous (kindred 2)	rare	P 33956074
p.Arg214Ter (R214*)	nonsense	compound-het with L204P	rare	P 42358996

- **Variant classification (ACMG/AMP):** the recurrent alleles are treated as pathogenic/likely pathogenic given segregation, absent/abolished protein, and functional data; many other *GIMAP5* entries in ClinVar are VUS or large 7q copy-number changes not specific to PHNC2.
- **Variant types:** predominantly **missense**; at least one **nonsense** (R214*). No pathogenic large structural rearrangement specific to PHNC2 is established (7q CNVs in ClinVar are contiguous-gene events, not PHNC2).
- **Functional consequence: loss of function** — e.g., compound-het L204P/R214 yields *absent GIMAP5 protein** on immunoblot ([P 42358996](#)).
No gain-of-function or dominant-negative mechanism.
- **Somatic vs germline:** exclusively **germline**.

Modifier genes. None formally proven. Mechanistically, GIMAP5 stability requires the lysosomal **MFSD1–GLMP** complex (

P 38055739

so *MFSD1/GLMP* are candidate biological modifiers; downstream **GATA4** and **GSK3β/CK2/ceramide-synthase** nodes modulate phenotype in models.

Epigenetic information / chromosomal abnormalities. No disease-specific DNA-methylation or histone signature reported. Contiguous 7q36 deletions/duplications affect *GIMAP5* among many genes but are not the mechanism of PHNC2.

5. Environmental Information

- **Environmental factors / toxins:** none required or established for PHNC2. (Toxic/drug-induced noncirrhotic portal hypertension — e.g., didanosine, thioguanine, arsenic, vinyl chloride, oxaliplatin — are separate acquired phenocopies to exclude.)
- **Lifestyle factors:** not applicable to disease causation.
- **Infectious agents:** not causal. However, because of the immune arm, patients suffer **recurrent/severe viral infections** as a *consequence* of the disease (e.g., severe viral infections in the proband,

P 42358996

CHEBI/pathogen exposure is downstream, not upstream.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. Biallelic **loss-of-function variants in *GIMAP5*** lead to loss/instability of the GIMAP5 GTPase (compounded by loss of the stabilizing lysosomal **MFSD1–GLMP–GIMAP5** complex). (*demonstrated*)

2. GIMAP5 loss **results in** failure to restrain **protein kinase CK2**, which **leads to** over-activation of **ceramide synthases** and **pathological accumulation of long-chain ceramides**. (*demonstrated in T cells; inferred in endothelium*)
3. Ceramide overaccumulation **leads to cellular senescence / organelle (mitochondrial) dysfunction** in GIMAP5-expressing cells — **endothelial cells and lymphocytes**. (*demonstrated*)

Branch A — hepatic vascular (portal hypertension): 4a. In **liver sinusoidal endothelial cells (LSECs)**, GIMAP5 loss **results in** reduced **GATA4** (the transcription factor required for LSEC specification). (*demonstrated by scRNA-seq positioning GIMAP5 upstream of GATA4*) 5a. Reduced GATA4 **leads to** loss of LSEC identity → **capillarization** (loss of fenestrae, acquisition of a basement membrane, **CD34 positivity**) and reduction of macrovascular hepatic endothelial cells. (*demonstrated in humans and mice*) 6a. Sinusoidal capillarization / porto-sinusoidal remodeling (with **nodular regenerative hyperplasia** and obliterative portal venopathy — venules absent in portal areas) **increases intrahepatic vascular resistance**. (*demonstrated histologically*) 7a. Increased resistance **results in noncirrhotic portal hypertension** → splenomegaly, esophageal varices, ascites, variceal bleeding; thrombocytopenia via hypersplenism. (*clinical*)

Branch B — immune (lymphopenia/autoimmunity): 4b. In lymphocytes, GIMAP5 loss (via ceramide, impaired mitochondrial/ER Ca^{2+} homeostasis, ER-stress/CHOP apoptosis, and constitutive **GSK3 β** activity restricting c-Myc/NFATc1) **leads to spontaneous T- and NK-cell apoptosis / impaired proliferation**. (*demonstrated in rat/mouse and patient cells*) 5b. Lymphopenia and immune dysregulation **result in recurrent infections, autoimmune cytopenias, T-cell exhaustion, and atypical memory B-cell expansion**. (*clinical*)

Key point (branch independence): The hepatic vascular disease is **cell-intrinsic to endothelium and lymphocyte-independent** — endothelial-specific *Gimap5* deletion reproduces capillarization, and *Gimap5^{sph/sph};Rag1^{-/-}* mice (lacking T/B cells) still develop the liver phenotype (

P 33956074

This predicts HSCT will not reverse established portal hypertension.

Category detail

- **Molecular pathways:** CK2 → ceramide-synthase/sphingolipid pathway (GO:0006672 ceramide metabolic process); **GATA4-dependent endothelial specification**; GSK3 β signaling (Wnt/insulin-adjacent) controlling c-Myc and NFATc1; NF- κ B/MAPK dysregulation

in *Gimap5*-mutant T cells

(
P 16584774
).

- **Cellular processes:** anti-apoptotic function (GO:0043066 negative regulation of apoptotic process); **cellular senescence** (GO:0090398); T-cell homeostasis (GO:0043029); endothelial cell differentiation (GO:0045446); autophagy/lysosomal biology.
- **Protein dysfunction:** loss of a small **AIG1-type G-domain GTPase** (Pfam PF04548; InterPro IPR006703; GO:0003924 GTPase activity; GO:0005525 GTP binding); missense variants in/ around the G-domain destabilize/abolish the protein (loss of function).
- **Metabolic/lipidomic changes:** accumulation of **long-chain ceramides** (CHEBI:17761 ceramide) — a defining metabolic signature.
- **Immune involvement:** combined features of immunodeficiency (lymphopenia, infection) and autoimmunity (cytopenias) — an inborn error of immunity.
- **Tissue-damage mechanism:** endothelial capillarization/senescence → sinusoidal fibrosis-independent remodeling; progressive liver dysfunction.
- **Molecular profiling:** single-cell RNA-seq of GIMAP5-deficient mouse liver documented replacement of LSECs by capillarized endothelial cells and placed GIMAP5 upstream of GATA4

(
P 33956074
);

T-cell proteomics/lipidomics documented the CK2–ceramide axis

(
P 38172257
).

GO / CL / CHEBI suggestions: GO:0003924, GO:0005525, GO:0043066, GO:0090398, GO:0006672, GO:0045446, GO:0043029; CL:1000488 (liver sinusoidal endothelial cell), CL:0000115 (endothelial cell), CL:0000084 (T cell), CL:0000623 (NK cell), CL:0000182 (hepatocyte); CHEBI:17761 (ceramide).

7. Anatomical Structures Affected

- **Organ level (primary): liver** (UBERON:0002107) — specifically the **hepatic sinusoid** (UBERON:0001281) and **portal venous system/portal vein** (UBERON:0002017). **Digestive/hepatobiliary and cardiovascular (portal venous) systems** are primarily involved.

- **Secondary organ involvement: spleen** (UBERON:0002106) — splenomegaly/hypersplenism; **esophagus** (UBERON:0001043) — varices; **bone marrow** (UBERON:0002371) and lymphoid organs — lymphopenia, extramedullary hematopoiesis (in models); gastrointestinal tract (variceal bleeding).
- **Tissue/cell level:** vascular **endothelium**, chiefly **liver sinusoidal endothelial cells** (CL:1000488); **T lymphocytes** (CL:0000084) and **NK cells** (CL:0000623); hepatocytes secondarily; platelets (consumed peripherally).
- **Subcellular level:** GIMAP5 localizes to the **lysosomal membrane** (GO:0005765) and **multivesicular body/endosome membrane** (GO:0032585) (UniProt Q96F15); older studies also implicated **ER** (GO:0005783) and **mitochondria** (GO:0005739) in T-cell survival/Ca²⁺ handling.
- **Localization / lateralization:** intra-abdominal, **bilateral/diffuse** hepatic involvement; not lateralized.

8. Temporal Development

- **Onset: congenital-to-pediatric;** portal hypertension and immune manifestations usually emerge in **infancy through adolescence**. Onset pattern is **insidious/chronic** (portal hypertension often detected via splenomegaly/thrombocytopenia or a variceal bleed).
- **Progression: chronic, progressive** hepatic vascular disease; can advance to decompensation (coagulopathy, hyperbilirubinemia) despite shunting
(

P 33956074

).
- Immune disease can be **episodic** (autoimmune cytopenia flares, intercurrent infections).
- **Disease course/duration: lifelong;** no spontaneous remission of the vascular disease. Immune cytopenias can be **treatment-induced remission** (e.g., sirolimus)
(

P 42358996

).
- **Critical periods / intervention windows:** early molecular diagnosis enables **variceal surveillance/prophylaxis**, infection prophylaxis, and consideration of HSCT before irreversible organ damage; the endothelial disease may have an early window before established capillarization/portal hypertension (inferred).

9. Inheritance and Population

- **Inheritance: Autosomal recessive** (HP:0000007). Recurrence risk 25% for siblings of an affected proband.
- **Penetrance:** appears **high/complete** for biallelic LOF (all reported biallelic individuals affected), though phenotypic **expressivity is variable** (severity, relative weight of hepatic vs immune features).
- **Genetic anticipation / mosaicism:** not applicable / not reported.
- **Consanguinity:** a **major factor** — most index families are consanguineous (

P 33956074

).
- **Founder / recurrent allele:** **p.Leu204Pro** (AF $\approx 2.1 \times 10^{-3}$) recurs and drives a disproportionate share of cases; no formal founder haplotype study published.
- **Carrier frequency:** *GIMAP5* is LOF-tolerant (gnomAD pLI=0.003); heterozygous carriers are healthy. Aggregate carrier frequency is low but non-trivial where p.L204P is present.
- **Epidemiology: ultra-rare** — on the order of ~20–25 **molecularly confirmed patients** reported worldwide (≥ 4 families in the founding study plus subsequent case reports/series, ~21 patients reviewed by 2026). No reliable prevalence/incidence estimate (well under 1/1,000,000). Reported patients are geographically diverse (Turkey, Europe, China, North America).
- **Sex ratio:** no established skew (autosomal recessive). **Age distribution:** pediatric-predominant at diagnosis.

10. Diagnostics

Genetic testing (definitive): - **Whole-exome sequencing (WES)** or **whole-genome sequencing (WGS)** identifying **biallelic *GIMAP5*** variants is the diagnostic gold standard (high-depth WES used in the founding study;

P 33956074

).

GeneMatcher/trio exome aids ultra-rare gene discovery. - **Single-gene / panel testing:** *GIMAP5* can be included on **PSVD / noncirrhotic portal hypertension** and **inborn errors of immunity** gene panels; targeted testing for the recurrent p.L204P is reasonable in consanguineous pedigrees. - **CMA/karyotype/FISH:** not primary; only relevant to exclude

contiguous 7q36 CNVs. Mitochondrial/repeat-expansion testing not applicable. - **Confirmatory functional test: GIMAP5 protein immunoblot** (absent protein confirms LOF) and **flow-cytometric immunophenotyping** (T/NK lymphopenia, T-cell exhaustion, atypical memory B cells)

(

P 42358996

).

Clinical/laboratory workup (supportive, to characterize the phenotype): - **Labs:** CBC (thrombocytopenia, cytopenias), transaminases (elevated), GGT (elevated), bilirubin/coagulation (with progression), lymphocyte subsets. LOINC-codable. - **Liver histology (PSVD pattern): nodular regenerative hyperplasia, CD34-positive capillarized LSECs, obliterative portal venopathy** (absent portal venules), **no cirrhosis/significant fibrosis**

(

P 33956074

).

SNOMED CT: nodular regenerative hyperplasia of liver. - **Endoscopy:** upper GI endoscopy for **esophageal/gastric varices**. - **Imaging:** ultrasound/Doppler, CT/MRI showing **splenomegaly, portosystemic collaterals, patent portal vein** (excluding portal/splanchnic vein thrombosis); transient elastography typically discordant with severe portal hypertension (low fibrosis vs high pressure).

Diagnostic criteria / differential. No PHNC2-specific criteria; diagnosis = PSVD/noncirrhotic portal hypertension features + biallelic *GIMAP5*. **Differential diagnosis:** cirrhosis of any cause; portal/splenic vein thrombosis (extrahepatic); congenital hepatic fibrosis; schistosomiasis; drug/toxin-induced NRH (didanosine, thioguanine, oxaliplatin); other genetic PSVD/INCPH genes — **DGUOK (PHNC1/INCPH1), KCNN3, FCHSD1, FOPV, TRMT5, HRG**, and syndromic causes (Adams–Oliver, telomere biology disorders, cystic fibrosis, Turner, Williams–Beuren) (review

(

P 38900412

).

GIMAP5's **accompanying immunodeficiency/autoimmunity** is a key distinguishing clue.

Screening (asymptomatic): cascade/carrier testing of relatives; prenatal/preimplantation testing in known families (see §13).

11. Outcome / Prognosis

- **Survival/mortality:** **guarded** in severe cases. In the founding cohort, three affected individuals in one kindred were deceased, **two dying from complications of portal hypertension**; one patient died at 17 with recurrent infections

(
P 33956074
).

No formal 5-/10-year survival statistics exist (too few patients).

- **Morbidity:** recurrent variceal bleeding, refractory/transfusion-dependent cytopenias, infection susceptibility, progressive liver dysfunction (coagulopathy, hyperbilirubinemia), and risk of extramedullary hematopoiesis (models). **Hepatocellular carcinoma (HP:0001402)** is listed as a potential long-term risk.

- **Recovery potential:** the vascular disease does not spontaneously remit; **portosystemic shunting improves varices** but does not halt underlying liver disease

(
P 33956074
).

Immune cytopenias can respond well to immunomodulation.

- **Prognostic factors:** degree of portal hypertension/hepatic decompensation; severity of immunodeficiency and infection burden; genotype (null/absent protein alleles such as R214 *may predict more complete deficiency*). *Prognostic biomarker candidate:** long-chain ceramide accumulation (mechanistic; not clinically validated).

12. Treatment

No disease-specific approved therapy exists; management is organ-directed and, increasingly, mechanism-informed.

A. Portal-hypertension / hepatic care (standard of care): - **Endoscopic variceal band ligation** and **nonselective beta-blockers** (e.g., propranolol/carvedilol) for variceal prophylaxis (NCIT: Propranolol C692; Endoscopic Variceal Ligation). - **Portosystemic shunt surgery / TIPS** — improved varices/collaterals in a reported patient

(
P 33956074
)

(NCIT: Transjugular Intrahepatic Portosystemic Shunt). - **Ascites management** (diuretics/sodium restriction); **liver transplantation** for end-stage disease (NCIT: Liver Transplantation C15356) — rational because the defect is intrahepatic-endothelial.

B. Immune-directed therapy: - **Sirolimus (rapamycin; mTOR inhibitor)** achieved **sustained remission** of **autoimmune** **cytopenias**

([P 42358996](#))

(NCIT: Sirolimus C1212). - Supportive: **IVIG**, targeted **antimicrobial/antiviral prophylaxis**, vaccination as tolerated (NCIT: Intravenous Immunoglobulin Therapy). - **Allogeneic hematopoietic stem cell transplantation (HSCT)** is considered **definitive for the immune/hematologic disease** (NCIT: Hematopoietic Stem Cell Transplantation C15431), but — because the liver disease is endothelial-intrinsic and lymphocyte-independent — **HSCT is not expected to reverse established portal hypertension**; the risk/benefit must be weighed carefully

([P 42358996](#))

C. Experimental / mechanism-based (preclinical): - **GSK3 inhibitors** rescued patient T-cell proliferation in vitro

([P 29382851](#))

- **CK2 inhibitors + ceramide-synthase inhibitors** rescued GIMAP5-deficient T cells by preventing ceramide overaccumulation

([P 38172257](#))

— a candidate for future targeted therapy. - **Gene/RNA therapy:** none developed; conceptually attractive given recessive LOF and endothelial/lymphoid expression (not yet in trials). No NCT registrations identified.

Pharmacogenomics: none disease-specific. **Personalized approach:** genotype/phenotype-guided balancing of liver-directed vs immune-directed (sirolimus) vs HSCT strategies.

13. Prevention

- **Primary prevention:** not possible for the genetic disease itself; **preconception genetic counseling** in consanguineous/known-carrier families, with **carrier testing, prenatal**

diagnosis, and **preimplantation genetic testing (PGT-M)** to prevent affected births (autosomal recessive, 25% recurrence).

- **Secondary prevention (early detection):** molecular diagnosis enables **variceal surveillance endoscopy**, monitoring of platelet counts/liver tests, and early immune monitoring; **cascade genetic screening** of at-risk relatives.
 - **Tertiary prevention (complication avoidance):** beta-blockers/band ligation to prevent variceal hemorrhage; infection prophylaxis and vaccination; timely immunomodulation for cytopenias; transplant planning.
 - **Immunization/public-health/environmental measures:** standard immunizations (with attention to live-vaccine caution in immunodeficiency); no vector/sanitation measures apply (non-infectious etiology).
 - **Counseling:** genetic counseling is central given recessive inheritance and consanguinity.
-

14. Other Species / Natural Disease

- **Taxonomy of affected/model species:** human *Homo sapiens* (NCBI:txid9606); *Mus musculus* (NCBI:txid10090); *Rattus norvegicus* (NCBI:txid10116). *GIMAP5* orthologs are also studied in **chicken** (*Gallus gallus*, NCBI:txid9031; PMIDs 31579581, 32147998).
 - **Orthologous genes:** mouse *Gimap5* (NCBI Gene 14468; MGI), rat *Gimap5* (RGD). Mechanisms (anti-apoptotic GTPase, lymphocyte survival) are **evolutionarily conserved** across rodents and birds.
 - **Natural (spontaneous) animal disease:** the **BioBreeding diabetes-prone (BB-DP) rat** carries a spontaneous *Gimap5* frameshift (*lyp*) allele causing **T-cell lymphopenia and autoimmune type 1 diabetes** (PMIDs 17655828, 19007993) — a naturally occurring model of GIMAP5 deficiency, though its hallmark is autoimmune diabetes rather than portal hypertension. No specific companion-animal PHNC2 (no OMIA entry for GIMAP5 portal hypertension identified).
 - **Comparative pathology:** rodents recapitulate lymphopenia and liver pathology; the **portal-hypertension/LSEC-capillarization** phenotype is specifically demonstrated in **mouse** (endothelial-intrinsic).
 - **Zoonotic potential:** none (genetic, non-transmissible).
-

15. Model Organisms

Mouse (*Mus musculus*; MGI) — **primary PHNC2 model:** - Germline *Gimap5* knockout / *sph* (**sphinx**) ENU allele: impaired peripheral T-cell survival, disrupted NK/NKT development, chronic hepatic hematopoiesis, **hepatocyte apoptosis and liver failure**, median survival ~15 weeks

([P 18796632](#)).

- **Endothelial-cell-conditional *Gimap5* deletion:** reproduces **LSEC capillarization** — the most faithful model of the human portal-hypertension mechanism

([P 33956074](#)).

- **Epistasis models:** *Gimap5^{sph/sph};Rag1^{-/-}* mice (no T/B cells) still develop liver disease → **portal hypertension is lymphocyte-independent**

([P 33956074](#)).

- **Complex/humanized biology:** *Mfsd1*, *Glmp*, and *Gimap5* germline knockouts each cause lymphopenia, liver pathology, and extramedullary hematopoiesis, defining the stabilizing **MFSD1–GLMP–GIMAP5** complex

([P 38055739](#)).

- **Mechanistic mouse models:** GSK3 β -axis studies

([P 29382851](#));

ceramide/CK2 senescence pathway

([P 38172257](#)).

Rat (*Rattus norvegicus*; RGD): BB-DP (*lyp/lyp*) and congenic lines — spontaneous *Gimap5* frameshift; robust model of **T-cell lymphopenia, ER-stress/CHOP apoptosis, mitochondrial Ca²⁺ dysregulation, and autoimmune diabetes** (PMIDs 17655828, 19424493, 19007993, 21502331).

In vitro / cellular: patient T cells (proliferation defect rescued by GSK3 inhibitors), Jurkat/HEK overexpression studies, and primary LSEC analyses.

Phenotype recapitulation & limitations: rodents faithfully model the **immune arm** (lymphopenia, autoimmunity) and, in mouse endothelial-specific/whole-body models, the **hepatic vascular arm** (capillarization, portal hypertension). **Limitations:** the BB-DP rat's dominant phenotype is autoimmune diabetes (not seen as a core human PHNC2 feature);

mouse whole-body nulls have very short lifespans complicating chronic-disease study; hepatocellular carcinoma risk and human-specific variant effects are not fully captured.
Resources: MGI (mouse *Gimap5*), RGD (rat *Gimap5*, BB rat), IMPC/IMSR for alleles.

Supported vs. Refuted Hypotheses

Supported: - PHNC2 is caused by **biallelic LOF *GIMAP5* variants, autosomal recessive** (P 33956074).
 - Portal hypertension arises from **GATA4-dependent LSEC capillarization, an endothelial-intrinsic, lymphocyte-independent** mechanism (P 33956074).
 - A unifying **CK2-ceramide senescence** pathway links hepatic and immune disease (P 38172257).
 - Disease is **multisystem** (portal hypertension + immunodeficiency/autoimmunity) with childhood onset (PMIDs 33956074, 42358996).

Refuted / not supported: - PHNC2 is **not** due to cirrhosis, fibrosis, or portal/splanchnic vein thrombosis (explicitly excluded; P 33956074).
 - Not caused by *DGUOK* (that is the distinct **PHNC1/INCPH1**, P 26874653).
 - No environmental/infectious cause is required; heterozygous carriers are unaffected.

Limitations and Future Directions

- **Very small N** (~20–25 patients) limits penetrance, expressivity, natural-history, prognosis, and QoL precision.
- **Genotype–phenotype** relationships (e.g., missense vs null alleles; the recurrent p.L204P; possible monoallelic vascular predisposition) need larger cohorts.

- **Therapeutic priorities:** validate CK2/ceramide-synthase and GSK3 inhibition clinically; define whether early HSCT alters liver outcomes; assess liver transplantation outcomes; explore endothelial-directed or gene-based therapy.
 - **Registry/omics:** a dedicated PSVD-genetics registry and hepatic single-cell/lipidomic profiling in patients would refine mechanism and biomarkers.
-

Key References (PMID)

- **33956074** Drzewiecki et al., *J Exp Med* 2021 — GIMAP5 maintains liver endothelial cell homeostasis and prevents portal hypertension (**disease-defining**).
 - **38172257** Park et al., *Nat Immunol* 2024 — GIMAP5 deficiency reveals a ceramide-driven longevity/senescence pathway.
 - **38055739** Zhong et al., *PNAS* 2023 — MFSD1–GLMP–GIMAP5 complex in lymphocyte survival and liver homeostasis.
 - **42358996** Moratti et al., *Front Immunol* 2026 — clinical spectrum, sirolimus, HSCT dilemmas (compound-het L204P/R214*).
 - **29382851** Patterson et al., *Nat Commun* 2018 — Gimap5–GSK3 β ; first human LOF patient.
 - **18796632** Schulteis et al., *Blood* 2008 — Gimap5-KO mouse liver failure/lymphopenia.
 - **38900412** Ciriaci et al., *Hepatology* 2026 — genetic predisposition to PSVD (differential/gene list).
 - **26874653** Vilarinho et al., 2016 — *DGUOK* in INCPH1 (distinct PHNC1).
-